Unit Testing is the process of checking small pieces of code to ensure that the individual parts of a program work properly on their own

- Unit testing done by developers.
- Unit tests are used to test individual blocks (Units) of functionality.

All basic units are tested before integration.

Integration Testing

is conducted to evaluate the compliance of a system or component with specified functional requirements. (Req written in SRS, is the product follows or not).
. It occurs after unit testing & before system testing.

System Testing :- is a level of testing that validates the complete & fully integrated software product.

→ Unit testing → Individual units are tested, Individual units ready → Integration testing performed → System Testing

Scanned with CamScanner

- The purpose of system test is to evaluate the end to end system specification. (SRS).
- System testing is a black box testing. (no internal structure details).
- System testing categories based on: Who is doing the testing?

<u>alpha testing</u>: Person develop the s/w check the complete s/w system

<u>beta testing</u>: Review of the s/w among different opinion.

<u>acceptance testing</u>: Customer / client, accept it or not,

~~System~~

System testing categories based on functional / Non functional Requirements.
(quality).

- this testing conducted on a complete integrated System to evaluate the system compliance with its specified requirements.

- Unit testing → Small components are tested → Done by development team.
- Small Components combine & integrated → Done integration testing done by development team.
- All components combine complete system ready → System testing

System testing is performed by a testing team that is independent of the development team that help to test the quality of system impartial.

- Black box testing → No knowledge of Internal structure, check the system.

Types of System testing

1. Performance testing :- is a type of s/w testing that is carried out to test the speed, scalibility, stability & reliability of the s/w product or application.

② **Load Testing** :- is a type of software testing which is carried out to determine the behaviour of system or s/w product under extreme load.

3 **Stress Testing** :- is a type of s/w testing performed to check the robustness of the system under varying loads. (Impose errors to check that how system react, so in real condition how system will behave).

4) **Scalability Testing** It is carried out to check the Performance of a s/w app. or system in term of its capability to scale up or scale down the no. of user requests load.

Internal structure knowledge needed in White Box testing, like Data structure, arrays, Control statements,

Black box testing no concern with internal structure, We just give input & check eitha we get required O/P properly.

White-box testing	Black box testing
The developers can perform white box testing.	The test engineers perform the black box testing.
what the software is supposed to do, also aware of how it does it.	what the software is supposed to do but is not aware of how it does it.
To perform WBT, we should have an understanding of the programming languages.	To perform BBT, there is no need to have an understanding of the programming languages.
In this, we will look into the source code and test the logic of the code.	In this, we will verify the functionality of the application based on the requirement specification.
In this, the developer should know about the internal design of the code.	In this, there is no need to know about the internal design of the code.
Test design techniques:Control flow testing, Data flow testing, Branch testing, Statement coverage, Decision coverage, Path testing.	Test design techniques: Decision table testing, All-pairs testing, Equivalence partitioning, Boundary value analysis, Cause–effect graph
Can be applied mainly at unit testing level but now in integration, system level also.	Can be applied virtually to every level of software testing: unit, integration, system and acceptance

White Box Testing

(glass box testing, transparent box testing, clear box testing or structural testing)

Is a method of s/w testing that tests internal structures or working of an application.

- Internal working/functionality/data flow is also checked.
- Example of white Box testing tools, Nunit, Veracode, CppUnit and RCUNIT etc.

White Box test design techniques for test Cases.

1) Control flow testing :- All the control flow statements are tested.

2) Data flow testing :- How data is flow ; (steps to perform a task).

3) Branch Testing :- Goto, Return, Break statements, must tested How control will move.

4) Statement Coverage : Every Statement must be tested.

5) Decision Coverage : Decision table / statement must be tested.

6) Path Testing :